

higher price after assembling. Presently, there is no such norm to limit the import of products and encourage indigenous manufacturing (PMA has included LED products but it is presently on hold). As a result products from China like MR16, strip lights and some downlights, which are cheaper and have better aesthetics, are more popular than the India-made products. The regulations that are present are also not being implemented strictly.

Lack of national standards. In absence of technical standards and norms, sub-standard, low-cost Chinese products have flooded the market, because of which indigenous manufacturers are suffering a lot. There are no takers for their high-quality but expensive products. Informing about the standards, which Indian manufacturers currently follow, Arun Gupta says, "There are certain standards by Bureau of Indian Standards (BIS) related to LED lamps and luminaires. The International Electrotechnical Commission (IEC) standards are the ones that manufacturers by and large follow."

Lack of testing. There is also lack of testing protocols, facilities and accredited laboratories at the national level, which also makes it difficult for the manufacturers.

Slow pace of government products. The major demands for the LED lighting products come from government departments but most of the government projects move at a very slow pace, which discourages the manufacturers.

Poor infrastructure and absence of a manufacturing ecosystem. Poor infrastructure, high capital costs, lack of a supply chain or vendor base and lower worker productivity have a direct bearing on the time to break even for manufacturers. There is also an absence of a capital goods industry, which could otherwise help in the automation of some processes and reduce dependence on operator skills.

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NEW TECHNOLOGY TRENDS

Due to the stiff competition from branded imported LED products, indigenous manufacturers focus more on improving the quality of their products and do a lot of R&D for advanced technologies. Thus it is not a surprise that, with respect to quality or technology, Indian LED lights are now on par with high-end, branded imported lights.

"One of the main trends in the LED segment is the miniaturisation of fixtures with increase in efficacy. AC LEDs are also getting prominence in certain products and applications. In addition, innovative cooling systems are being introduced to keep junction temperature in control, and efforts are also on to increase the efficacy of organic LEDs so that the same can be used for general lighting purposes," says Arun Gupta informing about the new technologies being introduced in the LED lighting sector.

Most of the companies are trying to improve the lumen-per-dollar ratio to make their products commercially viable, and for that they are going for low- and mid-power LEDs. Thus the high-power LEDs are being used only for industrial lighting products. This trend of shifting from high-power LEDs, which have unique features like higher junction temperature and increased luminous outputs, to low- and mid-power LEDs is because high-power LEDs require advanced knowhow of materials like aluminium, and costs increase in proportion to the size of the heat-sink used. With mid- and low-power LEDs, there are no such issues. Most of the

Chinese manufacturers also do not use good technology to make high-power LEDs to meet the increasing pressure of reducing the price of the end product.

Endnote

The trend of shifting from conventional light sources like high wattage incandescent bulbs and CFL to renewable energy products is catching up at a fast pace, thanks to the unique features of the controllable next-generation LED sources, which have control over the emission spectrum, polarisation, colour temperature and temporal modulation.

Though still imported products cater to the major chunk of the domestic market, encouragingly India-made products are also slowly gaining ground. There is a high possibility that the dependence on imported products will reduce significantly in the near future due to various factors like reliability, after-sales service, good quality of the domestic products, as well as the 2-3 years warranty coming with the India-made products. The demand for LED streetlights and LED solar lights is expected to grow rapidly in the coming years.

Now the prices of luminaires like downlights, bulbs and streetlights have come down to some acceptable level. 15W downlights, which were selling at ₹ 2500 each, are now selling at around ₹ 1500, whereas the price of LED tube-lights of 18W has come down from ₹ 3000 to around ₹ 1700. ●

The author is an assistant editor at EFY. Her report is based on inputs from Electronics Bazaar and a recent Frost & Sullivan report